Info type: Confidential
Company: NTT Data Payment Services India Limited
Info. owner: Product Management Group

[Callback API]

API Document

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Document Information

Document Attributes	Information
ID	New Callback API V1.03
Owner	Product Management C Engineering
Author	PCE - Product Team
Reviewer	Nital Kapadia

Document Version History

Version Name	Version Release Name
1.1	Old Paynetz Format in Open data Structure
1.2	Encrypted Json format without UDF parameters
1.3	User ID addition in UDF 5 (specific to Entab)
1.4	Encrypted Json format with UDF parameters
1.5	Real time OTS Data in Callback API for Null-Values
1.6	Country Code addition (specific to IRCTC)
1.7	Two additional parameters added under Customer Details

Contents

Introduction

This document provides an overview of call-back API process flow. Using the Callback API the merchant in which NTT Data payment services system posts transaction response info, without initiating any API request. It is only triggered when the transaction is successful or failed. This information is posted server-to-server using the POST method, in name value pair format on merchant's URL configured in NDPS system.

General process flow

The general process flow of the call-back API is as follows:

The end customer after purchasing product/service on merchant websites opts for payment, he either gets redirected to NTT Data payment service page or requested through API in case of seamless request.

The user selects his choice payment option and provides credentials for authorization.

Once the authorization is completed the transaction is executed. For a transaction to be successfully executed, the IP and domain URL need to be whitelisted and configured.

Once the end user completes transaction successfully and if callback URL is configured for that merchant, then the response will be posted on return URL as well as callback URL in both scenarios' success/failure.

Pre-requisite

Merchant's IP and domain URL need to be whitelisted and configured at NTT data payment system and NTT data payment server also needs to be whitelisted at their end.

Callback posting parameters

Parameter Name	Data Type (Length)	Conditional [CM]/ Optional[O]/ Mandatory [M]	Sample Value	Content/ Remarks
merchId	number (10)	M	446442	Unique ID assign by NDPS to merchant
merchTxnId	string (50)	M	“1234567890”	Unique transaction ID provided by merchant system
merchTxnDate	datetime (19)	M	"2023-07-13 20:46:00"	Transaction date must be in yyyy-MM-dd HH:mm:ss format
amount	number (12,2)	M	10.00	Total Transaction Amount (10 digits prior to decimal and 2 decimal)
atomTxnId	number (25)	M	2E+12	Max limit is of 25 digits
surchargeAmount	number (12,2)	O	0.00	surcharge amount(10 digits prior to decimal and 2 decimal)
prodName	string (50)	M	NSE	Product Name

prodAmount	number (12,2)	O	10.00	Product wise amount division as provided by merchant(10 digits prior to decimal and 2 decimal)
totalAmount	number (12,2)	M	10.00	Total amount [amount + surcharge amount] (10 digits prior to decimal and 2 decimal)
custAccNo	string(45)	O	“100000036600”	Customer account number
clientCode	string (45)	M	“12345”	As provided by merchant
txnInitDate	datetime (19)	M	"2023-07-13 20:46:00"	Transaction Initiation Date(yyyy-MM-dd HH:mm:ss)
txnCompleteDate	datetime (19)	CM	"2023-07-13 20:46:00"	Transaction Completion Date for all successful txn(yyyy-MM-dd HH:mm:ss)

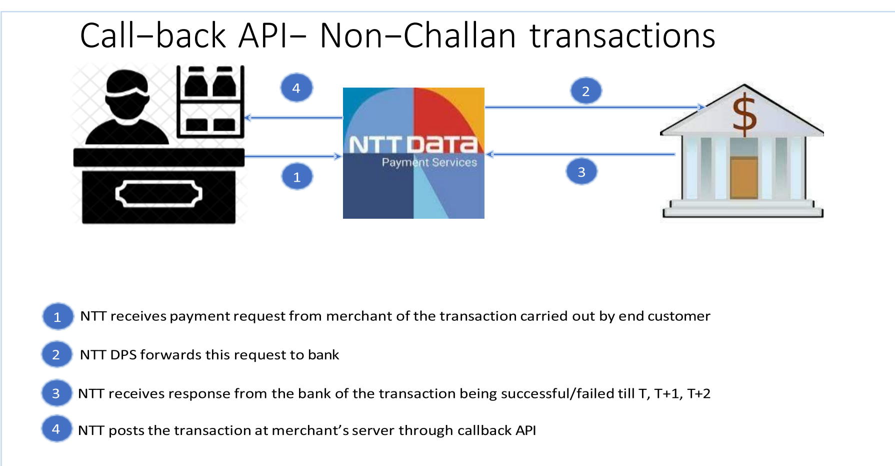
subchannel []	string (40)	M	"BQ"	Sub Channel (Payment Product will be displayed here)
otsBankId	string (10)	CM	"2"	Refer NB Bank List for possible values. Only applicable for NB, else null
otsBankName	string (60)	M	"HDFC Bank"	Bank Name
bankTxnId	string (30)	M	"UbiBgQp3Dwri0S347rge"	Bank Txn ID
cardType	string (20)	CM	"VISA"	Card Type will be presented for card transactions, else null
cardMaskNumber	string (20)	CM	"XXXXXXXXXXXX0465"	Mask Card Number will be presented card transactions, else null
statusCode	string (30)	M	"OTS0000"	Refer response codes
message	string (30)	O	"SUCCESS"	Message for Status Code
description	string (50)	O	"TRANSACTION IS SUCCESSFUL"	Status Description
udf1	string (45)	O		User Defined Parameter 1
udf2	string (45)	O		User Define Parameter 2

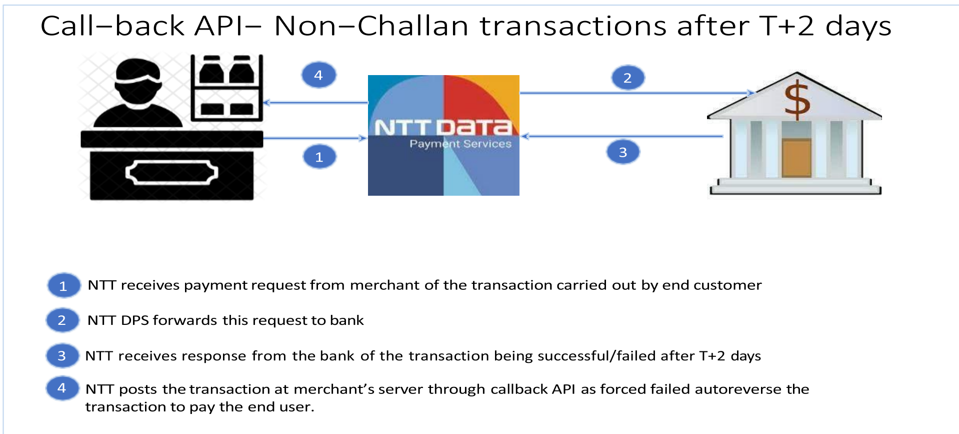
udf3	string (45)	O		User Define Parameter 3
udf4	string (45)	O		User Define Parameter 4
udf5	string (45)	O		User Define Parameter 5
udf6	string (255)	O		User Define Parameter 6
udf7	string (255)	O		User Define Parameter 7
udf8	string (255)	O		User Define Parameter 8
udf9	string (255)	O		User Define Parameter 9
udf10	string (255)	O		User Define Parameter 10

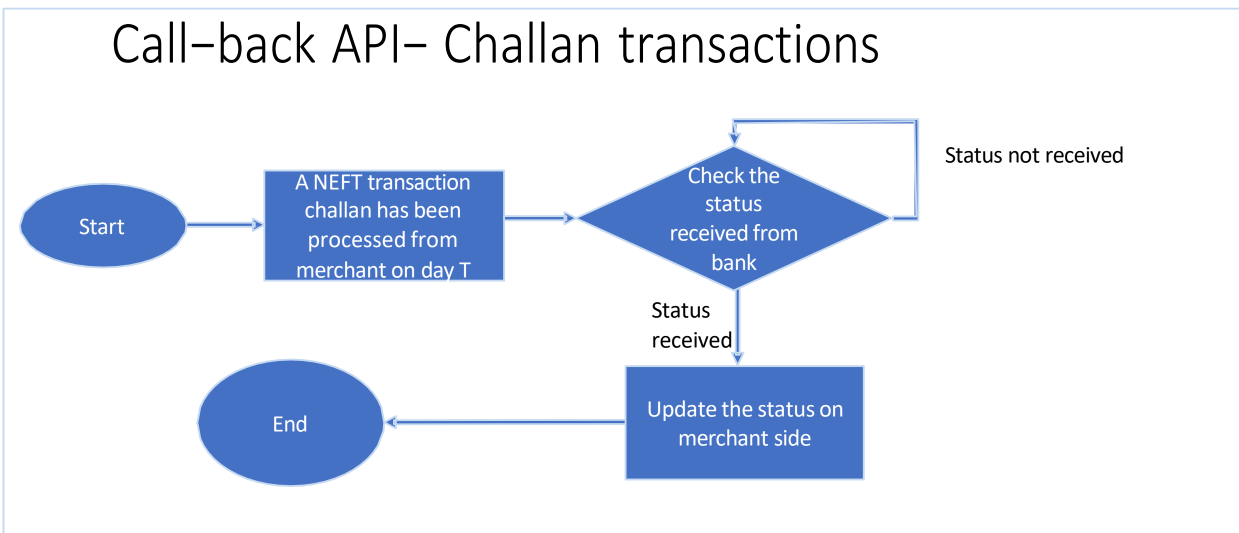
Please find details in form of attachment as well. :-

Request%20of%20Callback%20API_14022

Flow of call-back API







Sample Response Format

Currently, the response is sent in JSON format to the merchant server. The response is posted in JSON format.

Sample [Encrypted] Response

The encrypted form of the sample response which will be received by merchant will be as follows -

0CD63904F407DC4D24167F8D8F2BEE997C80D4536815B71C1EDDB8FDCBACE8B9780D809B0C
E258376F19B3F7E0411467D5B0B69EA8392D5B512407D2D77FBE35F9DFD026245247E6D81D64
F6708EBB3D93410AF13EAB9F10DEF033AEA6CFE1C321032E65E6DC8369AD7F442C444730C32
E81DA5DE8AE7EAAA69E45528A8DBF3C14F544DFECAFE84FE9DCB91D8E348D268B3E484C9298
CB233450B2C7147B2D6CFC1872B6B1DE6107DEFC26467C480F07217D66A6B35E569810CF9D7
49D86DDAEF66A589472F293B7544A0D7CE0EE726396C0387E01761D99F7340C3CB5B4CDE015
EA95EDB99AA9E718427313382A07A1CA5E991F1982980B2464C413746101121102 9BB90C0AC9
82A58258244501E0D023CC213C1F66E000780F3C5DC95F86273C8122A2A5EA121B69CF67415D
D0C13B14A6AD4045939FCA7D875E566787FBF6EF3CF0A73CC5F2D84F7B5092954566B2D1A24
910A057B13358DA7F91DA14C17E4ABFF24F72542FB025554FF7541F899B61AE2379F09B6022E2
3BABDC04417BFA1A6BB78F3FDD4B435F414771B36D5BB7843AEDCF99EADA3061DC96B2B961
7E8B12781818A037969EF5F28CC414DA66A0FD02E81A7251F6F4AF4F2443B09E4F4A38E42DC9
ECD9BE7474B82CFD4C53C69A95472C8CF56C39AB1173026FBEC6D72C9CCB6745DA3F571AE
C65C0CAA7E17E7AAEAC274696848F90360AED36E9F63B054630A9D312E2EF2AC8C87FF1854D
D5F8FAF7E14F3FCA8AE6F50A169BDFD56598C90220F4B3BC6DCCEE9BB4AC30C9DE47791588
B1C9F96379B478F505DDF3BD942626A52ED95A94E2307F0A871BA15B05B1178FFEE67F9F4FB2
338AEDB55E57651D294E1E6460B0B6CEF846BC011CA7A3C838DA9A8A01E1AAA1C1F04EB407A
F2345DEBF413B8053F80DBA888550B8EE1FF8DDFEB9B31B66BA879DE7B8BCAFCAABB274EDD
84CDE472A5FEFBDCD369815F55227DE7489B2A3DA7AA25609C1F06FC21C7785BBC4ADF42E4
1B1BE4B00DBBD5ED8CB5B3C20A1DD0A5A6A884A80C2627FFCBD933C9C7C0EB583DD873938
E4B620F36F184C16640E75E4D8C5CCE3F70D916002ABDEFF5B3DD14CCF0BC5061C34D56C94
B490751800B978AB9806F9913CE9F17C909F7F686F57E561DDB537DBBC9EDBB3AB8301C7B06
2DE01E4E3AEC75C9FD20A36B20D6AAF1B5F17D8F841EEC3B01E879C7B16124B216FA8567F9B
20A19530269EC549FDD3752BBBC8091106316516C1C233946D2C49D04ACFE3747647505D4F4
F07BE27812EC582FB6A935529B67F4F49EC4390E5B05E7DE62A0B327DC8ABD74B3760CF74C8
BC72E4059E4695BCD6E1764942204C5B439A2343EEA92E988981AD675AD159354642083DF450
721BE7500816B32992E962EF465E6D11E3AB2A4011D969BF13672A97F6ECC246FC499A685E821
5161D384EAF8619D5EA8008

Sample [Decrypted] Response

Below is the sample response:

{"payInstrument":
{"merchDetails":{"merchId":446442,"userId":null,"merchTxnId":"67a3157aa86f4","merchTxnDate":
"2025-02-05 08:38:34"},"payDetails":{"atomTxnId":11000000635176,"
prodDetails":[{"prodName":"NSE","prodAmount":1.00}],"amount":1.00,"surchargeAmount":0.00,"t

otalAmount":1.00,"custAccNo":null,"custAccIfsc":null,"clientCode":
"NAVIN","txnCurrency":null,"remarks":null,"signature":null,"txnInitDate":"2025-02-14
13:23:03","txnCompleteDate":"2025-02-14 13:24:11"},"payModeSpecificData"
:{"subChannel":["UP"],"bankDetails":{"otsBankId":2,"otsBankName":"Hdfc
Bank","bankTxnId":"1739519585428","authId":null,"cardType":null,"cardMaskNumber":null,
"scheme":null,"qrString":null,"tokenDetails":null},"cardDetails":{"cardScheme":"upi","cardMaskN
umber":null}},"extras":{"udf1":null,"udf2":null,"udf3":null,"u
df4":null,"udf5":null,"udf6":null,"udf7":null,"udf8":null,"udf9":null,"udf10":null},"custDetails":null,"
responseDetails":{"statusCode":"OTS0000","message":"S/P
UCCESS","description":"TRANSACTION IS SUCCESSFUL"}}}

Message Description:

In response message states pending and gets redirected in cases wherein response is awaited from the bank's side in case of dual verification. OTS0551 is the message code which gets reflected in case the message is received from the status is awaited from the bank's side.

Scenario 1-

If a transaction gets successful at NDPS' end, on real time NDPS will post "message = SUCCESS" response to merchant's Callback URL.

Scenario 2-

If on T-day, the transaction is pending at NDPS end, after bank re-query response on same day if transaction is success at bank's end then status will be changed from pending to success at NDPS' end, then NDPS will post "message =SUCCESS" response to merchant's Callback URL.

Scenario 3-

If on T-day, the transaction is pending at NDPS end, on T+1-day post reconciliation if transaction is success, then status gets changed from pending to success at NDPS' end, then NDPS will post "message =SUCCESS" (Force success) response to merchant's Callback URL. If it's Failed response will be "message =FAILED".

Scenario-4-

If on T-day, the transaction is getting failed at NDPS' end, on real time NDPS will post "message =FAILED" response to merchant's Callback URL.

Scenario-5-

Until and unless we are not getting response (Success/Failed) from bank end, NDPS will not trigger Callback API. Till T+2 if we're not getting response from Bank, NDPS system will mark transaction as Fail, but Callback will not be triggered. If transaction status is received as success from Bank's end after T+2, irrespective to the configuration [Force Success/Auto Reversal] in NDPS system, same will be marked as Auto Reversal.

Scenario-6-

For NEFT Transactions Challan Generated on T-day, in real time status will be posted on Merchant Return URL as ‘Pending’ with description ‘Challan generated successfully’.

Once end-customer makes the payment to the respective bank via challan, then the transaction status gets changed from ‘challan generated’ to ‘Success’ at NDPS end, then NDPS will post “message=SUCCESS” response to merchant’s Callback URL.

Response codes

Error Code	Message	Description
OTS0000	SUCCESS	TRANSACTION IS SUCCESSFUL
OTS0600	FAILED	TRANSACTION IS FAILED

Decryption of data

The data received by the merchant from NTT Data payment server will be in encrypted format using AES 512. To receive the data in JSON format, the data will have to be decrypted using response encryption credentials.

The sample key for UAT is -

MerchId	encResKey
446442	75AEF0FA1B94B3C10D4F5B268F757F11

Parameter Name	Mandatory	Data Type C Max Length	Sample Value	Content/ Remarks
encResKey	Mandatory	string(32)	75AEF0FA1B94B3C10D4F5B268F757F11	Need to be saved per MID as shared by NDPS

Java Code :

Java code for decryption

```
import java.util.logging.Logger;

import javax.crypto.Cipher;

import javax.crypto.SecretKey;

import javax.crypto.SecretKeyFactory;

import javax.crypto.spec.IvParameterSpec;

import javax.crypto.spec.PBEKeySpec;

import javax.crypto.spec.SecretKeySpec;

public class AtomEncryption

{

  static Logger log = Logger.getLogger(AtomEncryption.class.getName());    private static int pswdIterations = 65536;

  private static int keySize = 256;    private static final byte[] ivBytes = { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 };

  public static String decrypt(String encryptedText, String key)

  {  try

  {

byte[] saltBytes = key.getBytes("UTF-8"); byte[] encryptedTextBytes = hex2ByteArray(encryptedText);

SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA512");

PBEKeySpec spec = new PBEKeySpec(key.toCharArray(), saltBytes, pswdIterations, keySize);

SecretKey secretKey = factory.generateSecret(spec); SecretKeySpec secret = new SecretKeySpec(secretKey.getEncoded(), "AES");

 IvParameterSpec localIvParameterSpec = new IvParameterSpec(ivBytes);  Cipher cipher =
Cipher.getInstance("AES/CBC/PKCS5Padding"); cipher.init(2, secret, localIvParameterSpec);
```

```
byte[] decryptedTextBytes = (byte[])null;   decryptedTextBytes = cipher.doFinal(encryptedTextBytes);

 return new String(decryptedTextBytes);

  }

catch (Exception e) {

log.info("Exception while decrypting data:" + e.toString());

  }

return null;

}

 private static byte[] hex2ByteArray(String sHexData) {

   byte[] rawData = new byte[sHexData.length() / 2];

   for (int i = 0; i < rawData.length; ++i) {

     int index = i * 2;

     int v = Integer.parseInt(sHexData.substring(index, index + 2), 16);

     rawData[i] = (byte)v;

   }

   return rawData;

 }

 public static void main(String[] args) {

try {

String decryptedData = AtomEncryption.decrypt("Encdata", "ASWKLSLLFS4sd4g4gsdg");

System.out.println("decryptedData : "+decryptedData);

} catch (Exception e) {

// TODO: handle exception

}

}

}
```

UAT environment details.

Merchant Id	446442
Transaction Password	Test@123
product Id	NSE
Request Hash Key	KEY123657234
Response Hash Key	KEYRESP123657234
Encryption Key	A4476C2062FFA58980DC8F79EB6A799E
Decryption Key	75AEF0FA1B94B3C10D4F5B268F757F11
MCC Code	5999
API URL	https://paynetzuat.atomtech.in/ots/payment/txn